

$\mathbb{Z} = \{ \dots, -2, -1, 0, 1, 2, \dots \}$ Integers / Heltal

$\mathbb{Z}^+ = \{ 1, 2, 3, \dots \}$ positive tal

$\mathbb{Z}^- = \{ \dots, -3, -2, -1 \}$ negative tal

$\mathbb{N} = \{ 0, 1, 2, \dots \}$ Naturlige tal

$\mathbb{Q} = \{ \frac{m}{n} \mid m \in \mathbb{Z}, n \in \mathbb{Z}^+ \}$ rationell tal

$\mathbb{R} =$ Reelle tal

$\emptyset = \{ \}$ den tomme mængde

$E = \{ 2, 4, 6, 8, \dots \} = \{ n \in \mathbb{Z} \mid \exists k \in \mathbb{N} : n = 2k \} = \{ 2k \in \mathbb{Z} \mid k \in \mathbb{N} \}$

$O = \{ 1, 3, 5, 7, \dots \} = \{ n \in \mathbb{Z} \mid \exists k \in \mathbb{N} : n = 2k+1 \} = \{ 2k+1 \in \mathbb{Z} \mid k \in \mathbb{N} \}$

countable

finite = $\{ 1, 2, 3 \}$

countable infinite = $|\mathbb{N}| = \aleph_0$

uncountable

$|\mathbb{R}| = \aleph_1$

$\subseteq$: subset

$\subset$: proper subset

$|P(A)| = 2^{|A|}$

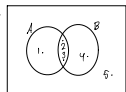
$A = \{ 1, 2, 3 \}$ $B = \{ 2, 3, 4 \}$ $U = \{ 1, 2, 3, 4, 5 \}$

$A \cap B = \{ 2, 3 \}$

$A \cup B = \{ 1, 2, 3, 4 \}$

$A \setminus B = \{ 1 \}$

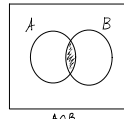
$\bar{A} = U \setminus A = \{ 4, 5 \}$



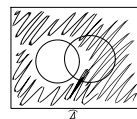
De Morgans Law for Sets

$$\overline{A \cup B} = \bar{A} \cap \bar{B} \text{ and } \overline{A \cap B} = \bar{A} \cup \bar{B}$$

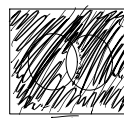
$\bar{A} \cup \bar{B}$



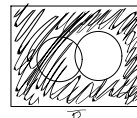
$A \cap B$



$\bar{A}$



$\bar{A} \cap \bar{B}$



$\bar{B}$

Tuples

$(1, 3, 2, 1) \neq (1, 3, 2)$

$(1, 3, 2) \neq (1, 2, 3)$

3 occurs in $(1, 2, 3)$

~~$5 \in (1, 2, 3)$~~

$\{1, 2, 3\} \subseteq \{1, 2, 3\}$

Subset

$\{3\} \subset \{1, 2, 3\}$

Proper Subset

$\{7, 8\} \not\subseteq \{1, 2, 3\}$

Not a Subset

$2 \in \{1, 2, 3\}$

Element

$5 \notin \{1, 2, 3\}$

Not an element

Cantors diagonalization:

Assume towards a conclusion that $\mathbb{R}$ is countable.
Consider the set $R_{0.1}$ of all reals with a 0 before the decimal point and any sequence of 0's and 1's after the decimal point.

Since $R_{0.1} \subseteq \mathbb{R}$, $R_{0.1}$ is also countable, so there exists an enumeration

$1 \rightarrow 0.d_{11}d_{12}d_{13}d_{14} \dots$ $0.1-d_{11}1-d_{12}1-d_{13} \dots$

$2 \rightarrow 0.d_{21}d_{22}d_{23}d_{24} \dots$

$3 \rightarrow 0.d_{31}d_{32}d_{33}d_{34} \dots$

we construct number s by inverting diagonal, i.e.,

$s = 0.d_1d_2d_3d_4 \dots$ where $d_i = 1-d_{ii}$ for all i . Then $s \in R_{0.1}$, but s does not occur in the list. Indeed, suppose that s occurs in position i in the list.

proof by contradiction

